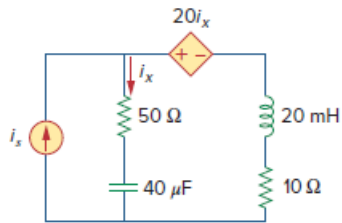


Problem

For the circuit in Fig. 11.38, $i_s = 6 \cos 103t$ A. Find the average power absorbed by the $50\text{-}\Omega$ resistor.

Figure 11.38:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

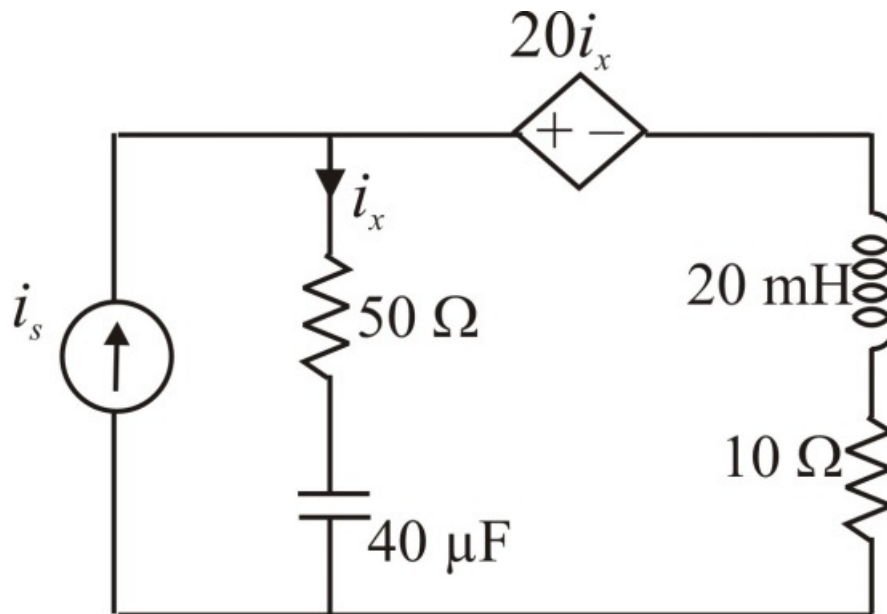


Figure 1

Step 2 of 5

Convert the circuit to the frequency domain:

$$\begin{aligned}i_s &= 6 \cos 10^3 t \text{ A} \\ &= 6 \angle 0^\circ\end{aligned}$$

And

$$\omega = 10^3 \text{ rad/s}$$

Convert the inductance value.

$$\begin{aligned}L &= 20 \text{ mH} \\ j\omega L &= j(10^3 \text{ rad/s})(20 \text{ mH}) \\ &= j(10^3)(20)(10^{-3}) \\ &= j20 \Omega\end{aligned}$$

Convert the capacitance value.

$$\begin{aligned}C &= 40 \mu\text{F} \\ \frac{1}{j\omega C} &= \frac{1}{j(10^3 \text{ rad/s})(40 \mu\text{F})} \\ &= -j25 \Omega\end{aligned}$$

Step 3 of 5

Draw the equivalent frequency domain circuit:

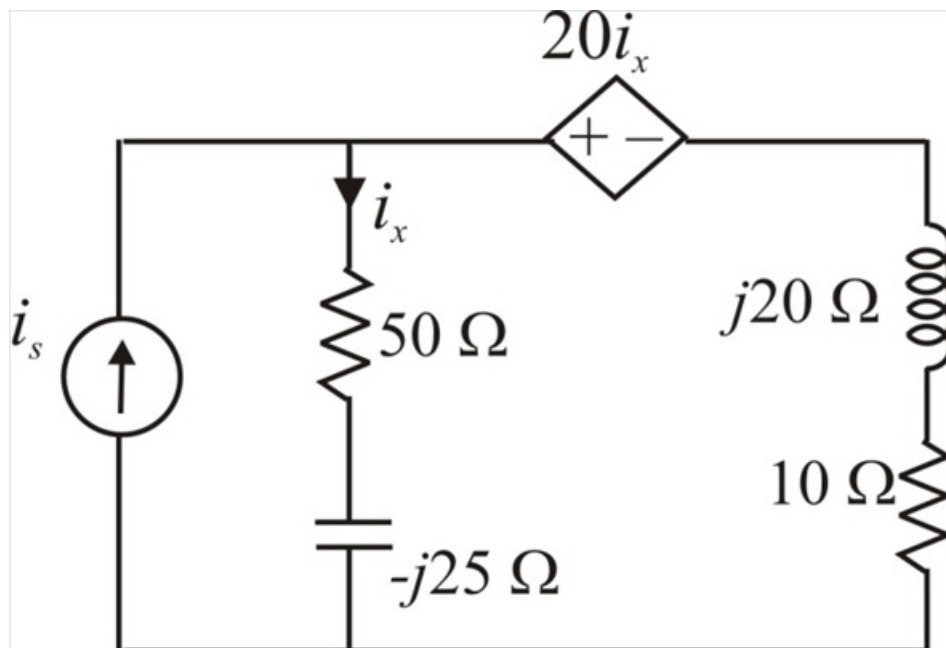


Figure 2

Step 4 of 5

Apply Kirchhoff's current law at node v_o :

$$\frac{V_o}{50 - j25} + \frac{V_o - 20i_x}{10 + j20} = 6$$

The following is the value of i_x from the circuit.

$$i_x = \frac{V_o}{50 - j25}$$

Therefore,

$$\begin{aligned} 6 &= \frac{V_o}{50 - j25} + \frac{V_o}{10 + j20} - \left(\frac{20}{10 + j20} \right) \left(\frac{V_o}{50 - j25} \right) \\ &= V_o (0.016 + j0.008 + 0.02 - j0.04 - 0.0128 + j0.0096) \end{aligned}$$

$$\begin{aligned} V_o &= \frac{6}{0.0232 - j0.0224} \\ &= 186.05 \angle 43.99^\circ \end{aligned}$$

Calculate the value of i_x :

$$\begin{aligned} i_x &= \frac{V_o}{50 - j25} \\ &= \frac{186.05}{55.90} \\ &= 3.328 \end{aligned}$$

Calculate the rms value of i_x :

$$\begin{aligned} i_{\text{rms}} &= \frac{i_x}{\sqrt{2}} \\ &= \frac{3.328}{\sqrt{2}} \\ &= 2.353 \text{ A} \end{aligned}$$

Step 5 of 5

Calculate the average power absorbed by the 50Ω resistor:

$$\begin{aligned} P_{\text{avg}} &= i_{\text{rms}}^2 R \\ &= (2.353 \text{ A})^2 (50 \Omega) \\ &= (2.353)^2 (50) \\ &= 276.8 \text{ W} \end{aligned}$$

Therefore, the power absorbed by 50Ω resistor is 276.8 W .